

GEOGRAPHY

PHYSICAL GEOGRAPHY



MODULE - 15

- CONTINENTAL DRIFT
- MID-OCEANIC RIDGES
- MID-ATLANTIC RIDGE



ALFRED WEGENER



WHAT DO YOU UNDERSTAND BY CONTINENTAL DRIFT?

- 1 It is the large-scale horizontal movement of continents relative to one another and to the ocean basins during one or more episodes of geologic time.
- 2 This is important precursor to the development of the theory of plate tectonics.
- 3 In 1912, German Meteorologist Alfred Wegener presented the concept of Continental Drift.
- 4 As per Wegener, during most of the geologic time, there was only one continent, which he called "Pangea"

CONTINENTAL DRIFT OF PLATES



225 Million Years Ago



150 Million Years Ago



100 Million Years Ago



Earth Today



MORE ON THE CONTINENTAL DRIFT



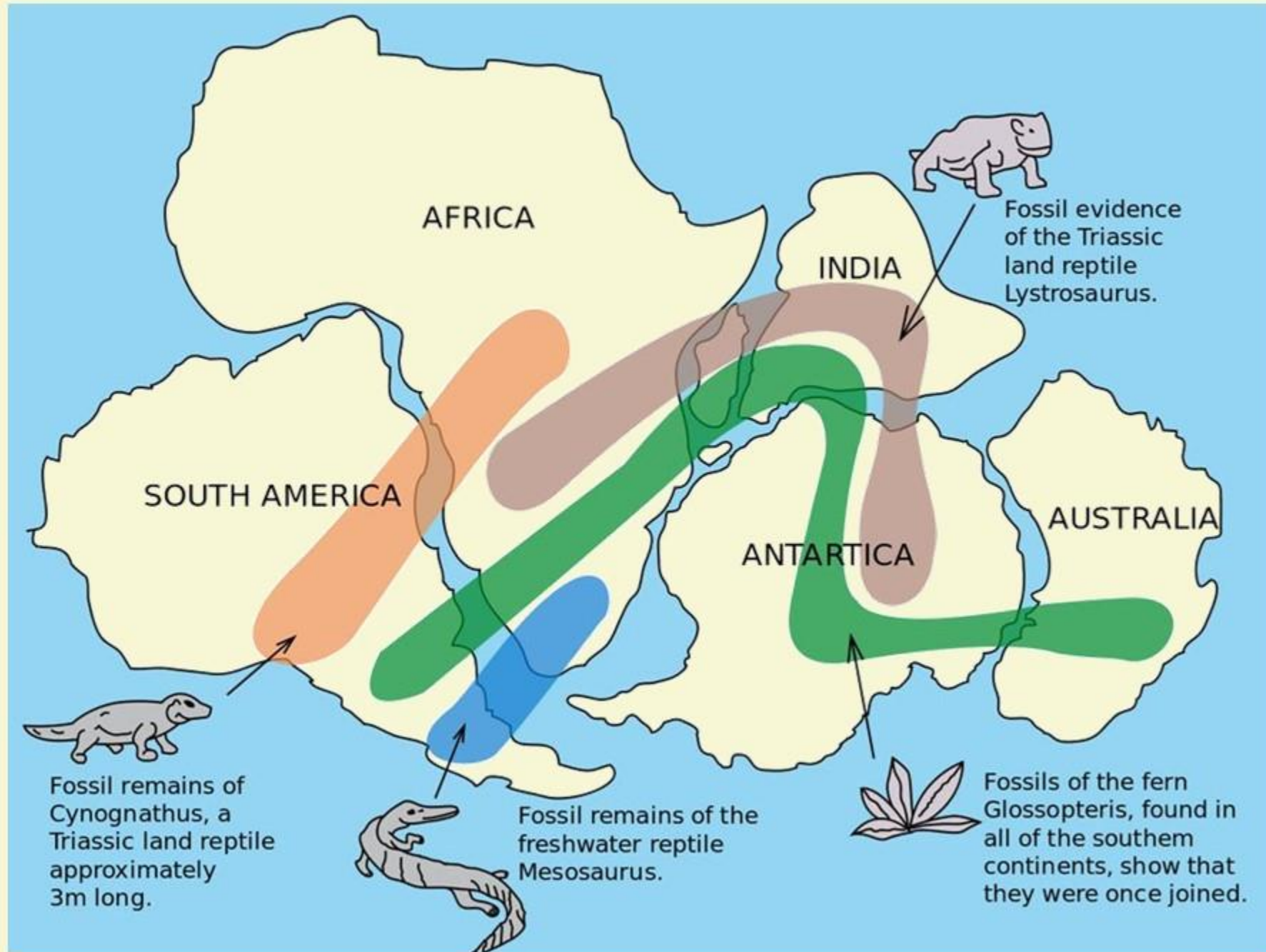
ALFRED WEGENER

1 Wegener published a paper explaining his theory that the continental landmasses were drifting across the Earth, sometimes plowing through the Oceans and into each other. He called this movement “Continental Drift”.

2 He felt that the Earth’s continents were once part of an enormous, single landmass called Pangea.

3 He came to the conclusion, because fossils of the ancient reptile mesosaurus are found in Southern Africa and South America.

4 The presence of mesosaurus in both the present continents suggest a single habitat with many lakes and rivers.





MORE ON THE CONTINENTAL DRIFT

5 He also came to the conclusion, because of the fossils from the frigid Arctic archipelago of Svalbard, Norway.

- These plants were not the hardy specimens adapted to survive in Arctic climate.
- These are tropical plants, which are adapted to a much warmer and more humid environment. So the presence of fossils suggest Svalbard once had a tropical climate.

6 He also studied the stratigraphy of different rocks and mountain ranges.

- The East Coast of South America and the West Coast of Africa seems to fit together
- Wegener discovered that the Appalachian Mountains of eastern United States were geologically related to the Caledonian Mountains of Scotland.

7 Pangea existed about 240 million years ago. Over millions of years, Pangea separated into pieces that moved away from one another.



**SVALBARD
GLOBAL SEED VAULT**

NORWAY

NORWEGIAN
ARCHIPELAGO

GREENLAND

Greenland Sea

Barents Sea

Norwegian Sea

SWEDEN

FINLAND

RUSSIA

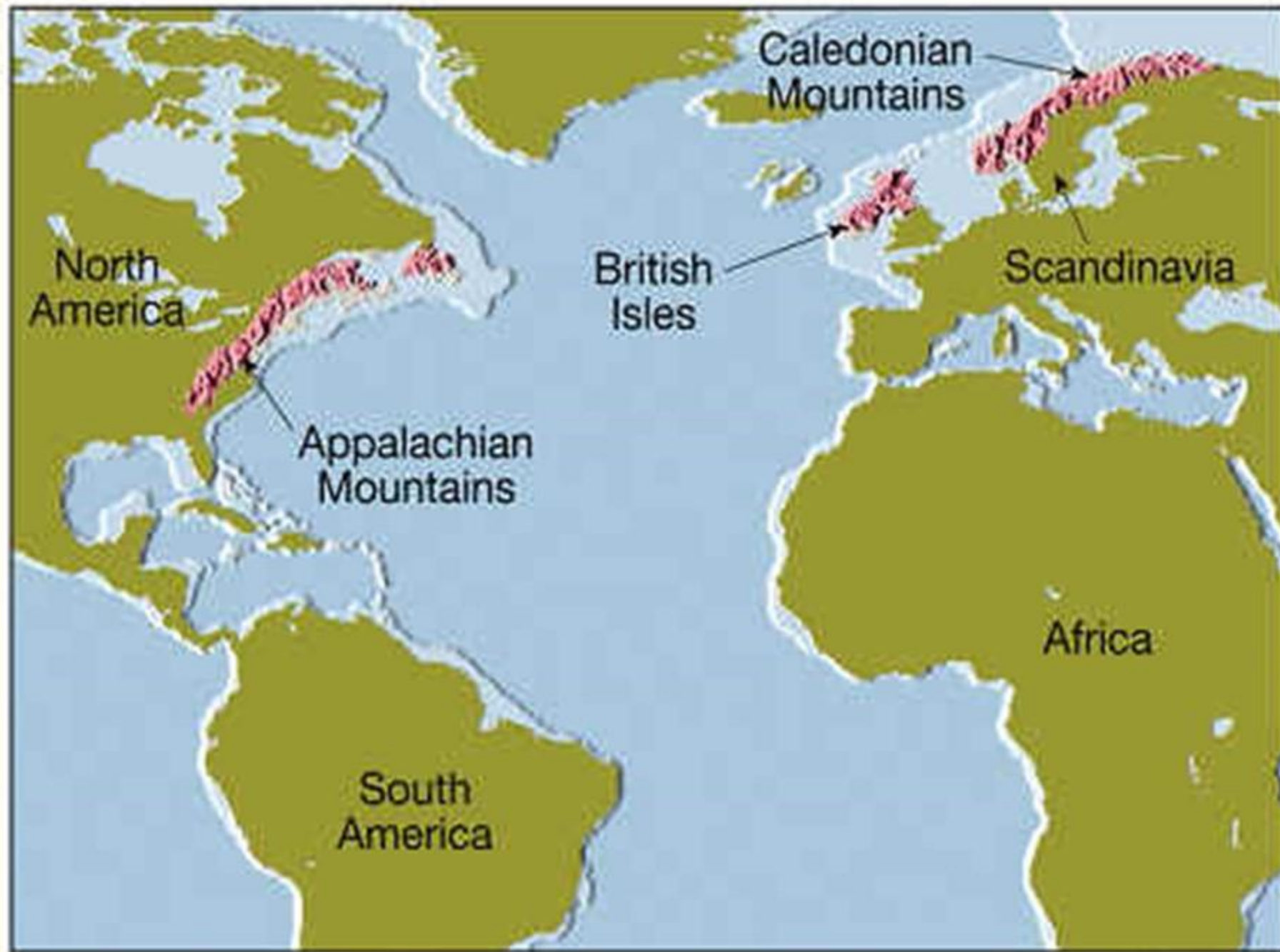
ICELAND

White Sea

Gulf of Bothnia

Baltic Sea

North



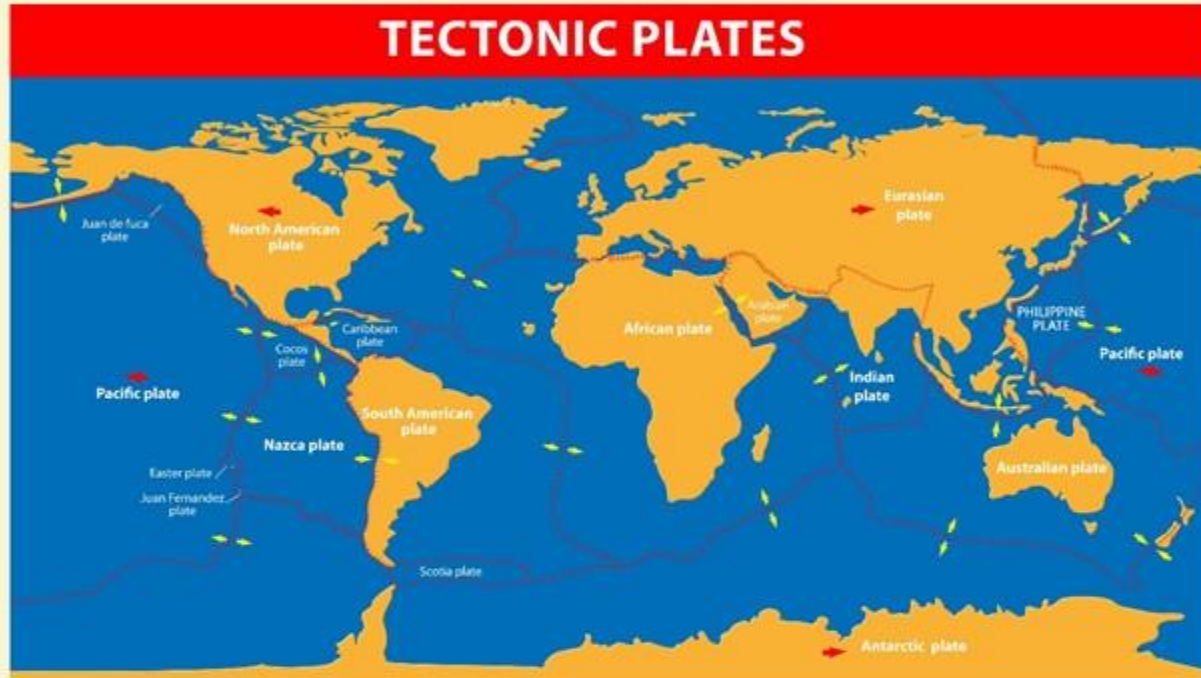


WHAT IS THE VIEWPOINT OF THE PRESENT DAY SCIENTISTS?

- 1 Today Scientists think that several supercontinents like Pangea have formed and broken up over the course of the Earth's lifespan.
- 2 One of the elements lacking in the theory was, the mechanism for how it works, why the continents drift and what patterns did they follow?
- 3 Wegener suggested that perhaps the rotation of the Earth caused the continents to shift towards and apart from each other.
- 4 Today, we know that the continents rest on a massive slabs of rock called Tectonic Plates.
- 5 The plates are always moving and interacting in a process called Plate Tectonics.

CONTINENTAL DRIFT





WHAT DO YOU UNDERSTAND BY PLATE TECTONICS?

- 1 Tectonic activity describes the interaction of the huge slabs of lithosphere, called the Tectonic Plates.
- 2 Earth's lithosphere is known for tectonic activity.
- 3 Tectonic activity is the movement of tectonic plates resulting in geologic activity such as volcanic eruptions and earthquakes.

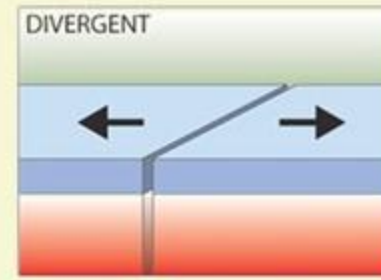
The lithosphere is divided into 15 major tectonic plates—the North American, Caribbean, South American, Scotia, Antarctic, Eurasian, Arabian, African, Indian, Philippine, Australian, Pacific, Juan de Fuca, Cocos, and Nazca.



PLATE TECTONICS

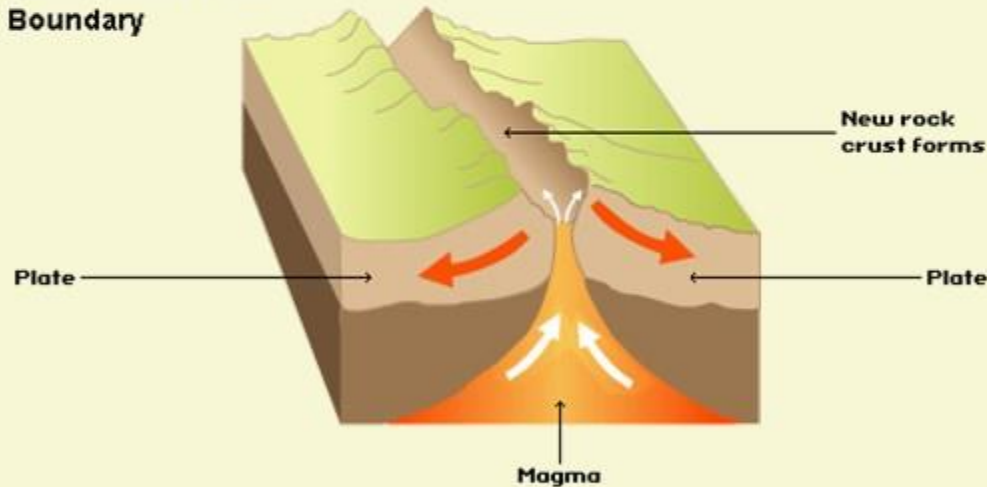
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CONSTRUCTIVE / DIVERGENT BOUNDARY



LECTURE OF MODULE NUMBER-14

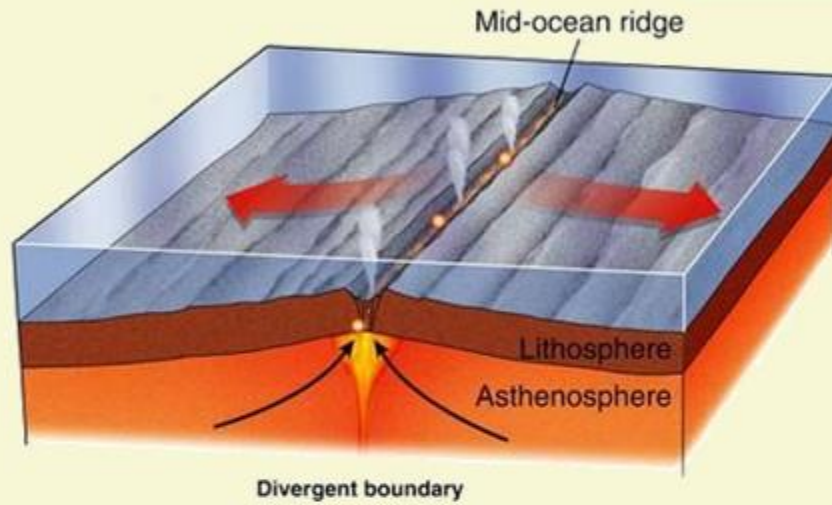
Constructive Plate Boundary



- 1 At divergent boundaries in the oceans, magma from deep in the Earth's mantle rises towards the surface and pushes apart two or more plates.
- 2 Mountains and volcanoes rise along the seam.
- 3 The process renews the ocean floor and widens the giant basins.
- 4 A single mid-ocean ridge system connects the world's oceans, making the ridge the longest mountain range in the world.
- 5 One example of the Continental Plates moving apart is Great Rift Valley in Africa.

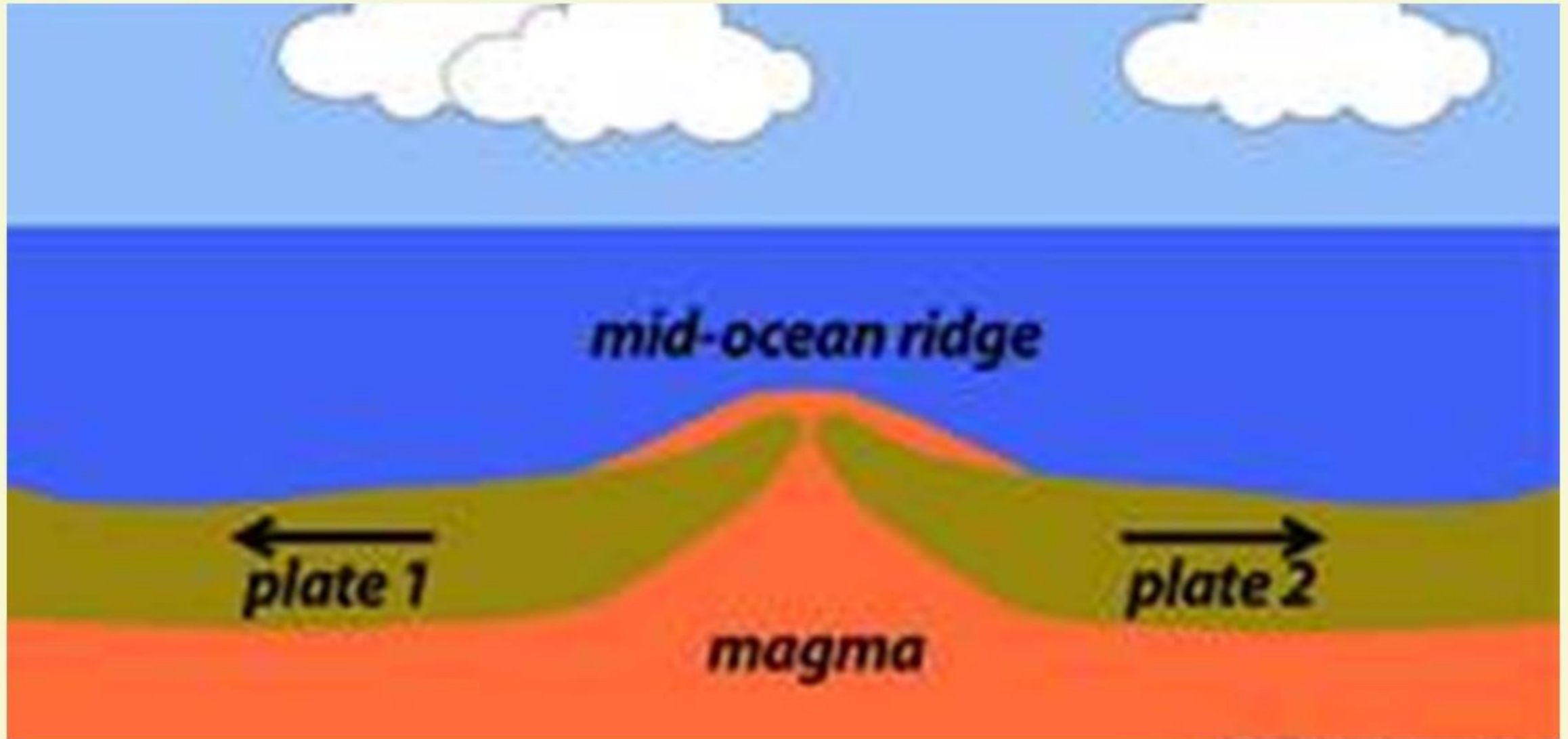
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CONSTRUCTIVE / DIVERGENT BOUNDARY

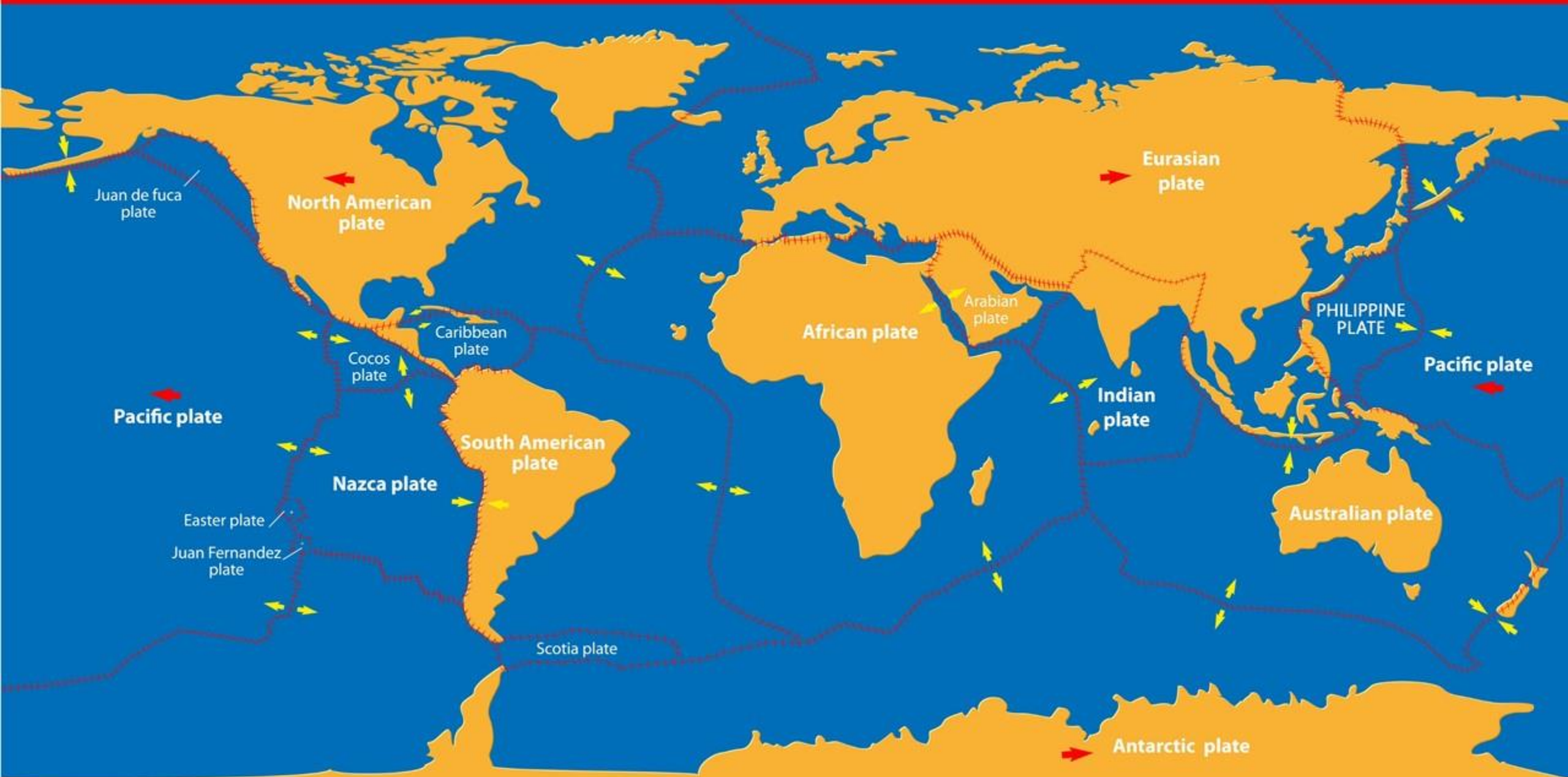


MID-OCEAN RIDGE

- 1 In the process of seafloor spreading, molten rock rises from within the Earth and adds new seafloor to the edges of the old.
- 2 Seafloor spreading is most dynamic among giant underwater mountain ranges known as Mid-Ocean Ridges.
- 3 As the seafloor grows wider, the continents move away from each other.
- 4 The North American and Eurasian tectonic plates are separated by Mid-Atlantic Ridge.

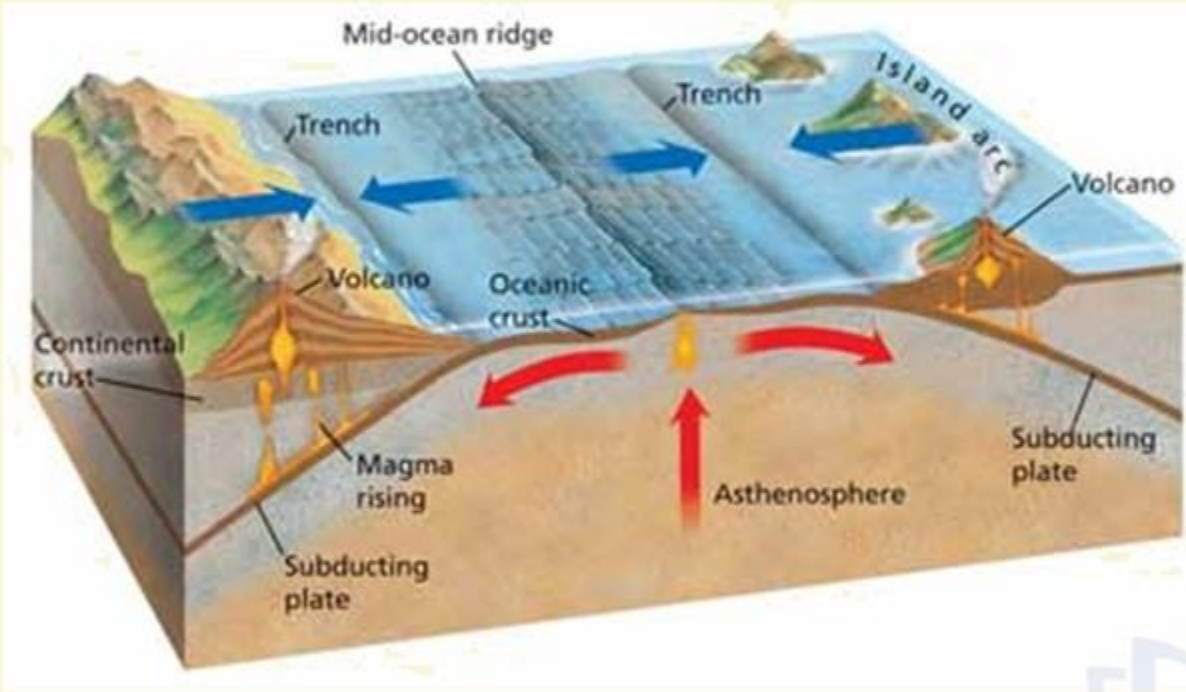


TECTONIC PLATES



1

CONSTRUCTIVE / DIVERGENT BOUNDARY

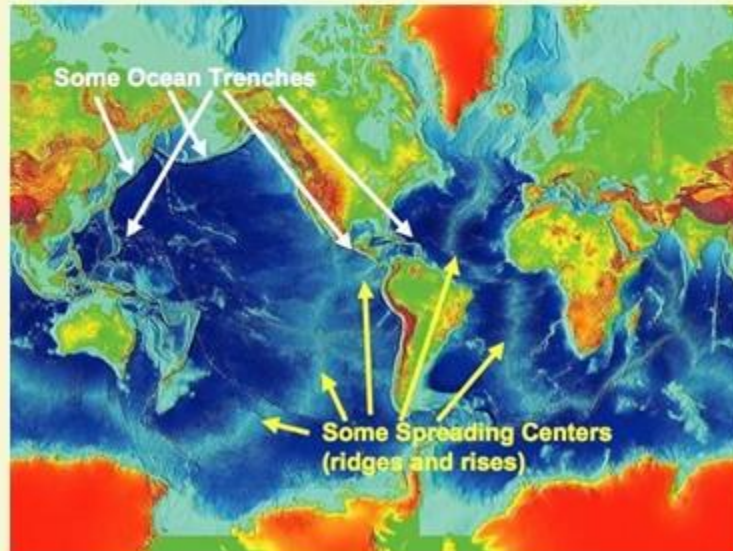
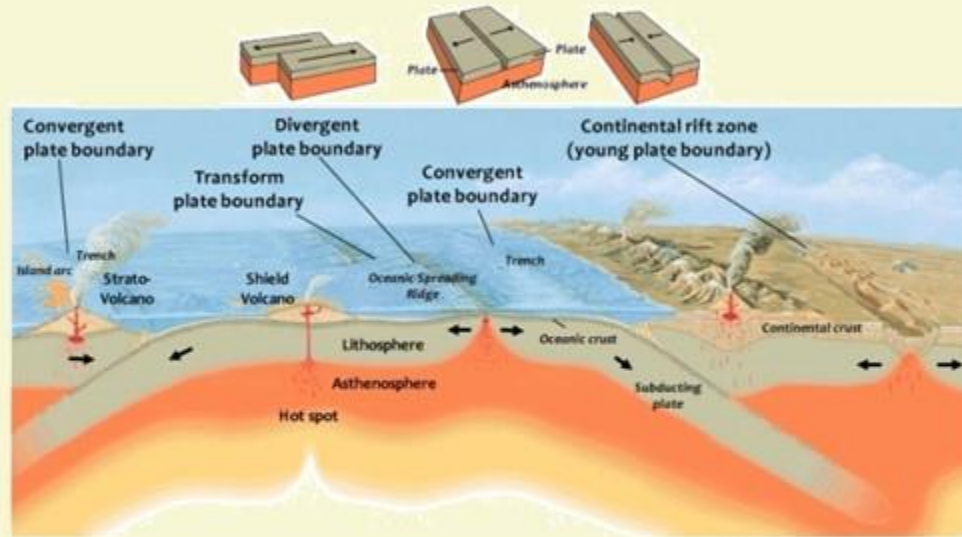


MORE ON MID-OCEAN RIDGE

- 1 It is the most extensive chain of mountains on Earth.
- 2 More than 90% of these mountain ranges lie in the deep ocean.
- 3 The Mid-Ocean Ridge runs around the Globe for more than 65,000 km.
- 4 It has an average water depth to the top of the ridge of 2,500 meters. It raises above sea level in Iceland and some other places.

1

CONSTRUCTIVE / DIVERGENT BOUNDARY

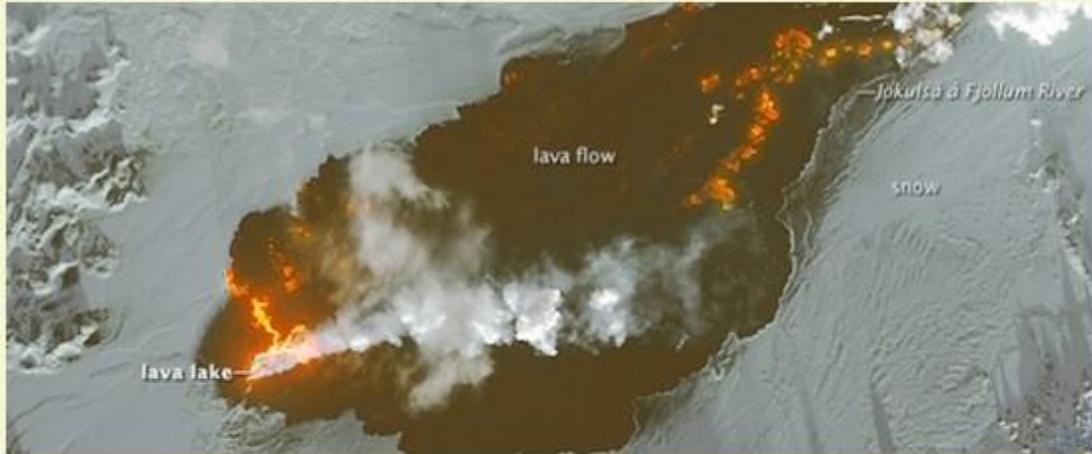


WHY MID-OCEAN RIDGES ARE IMPORTANT?

- 1 These are geologically important because at the plate boundary new ocean floor is created as the plates move apart.
- 2 It is the spreading centre or the divergent plate boundary or constructive plate boundary.
- 3 As the oceanic plates move apart, the molten rock moves up from tens of kilometers deep.
- 4 Some of the molten rock ascends all the way up to the sea floor producing enormous volcanic eruptions of basalt. It is building the longest chain of volcanoes in the world.

1

CONSTRUCTIVE / DIVERGENT BOUNDARY



WHAT HAPPENED IN 1783?

1

Normally the molten rock does not erupt and it freezes on the edges of the plates as they spread apart.

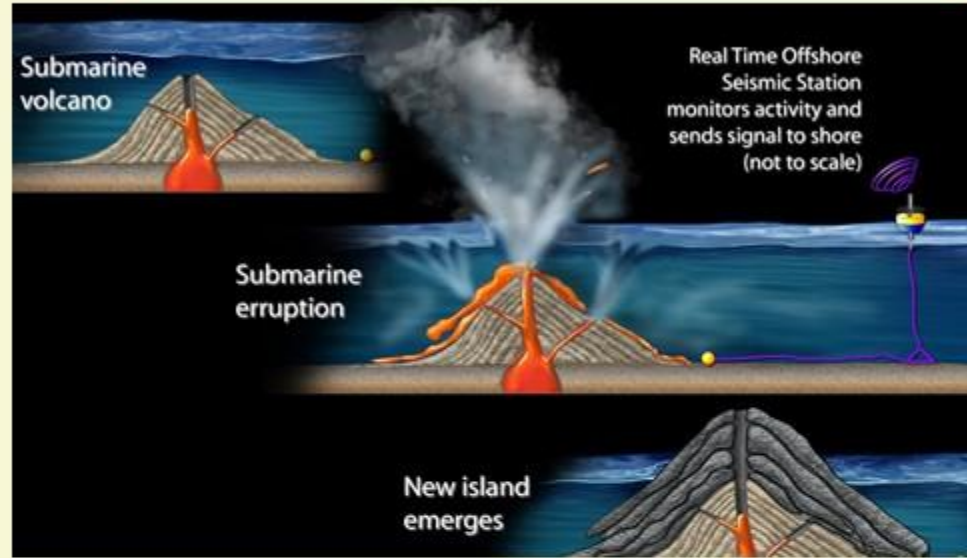
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In 1783, a segment of the Ridge, which emerges above sea level in Iceland erupted more than 12 cubic km of lava, which was at 1200°C as well as 50 million tonnes of Sulphur-dioxide. It caused death of around 10,000 Icelanders, a quarter of the Nation's population at that time.

Since most of the Mid-Ocean Ridge is more than 2,000 mts deep, most of the eruptions go unnoticed.

1

CONSTRUCTIVE / DIVERGENT BOUNDARY



HOW THE MID-OCEANIC RIDGE LOOKS LIKE?

- 1 The zone of recent volcanic activity is narrow and less than 10 kms and often even less than 1 km wide.
- 2 This region generating new ocean floor is characterized by many small to moderate earthquakes.
- 3 Some of the earthquakes are caused by volcanic eruptions, others are caused by breaking and ripping of the thin newly created plate, as it spreads to either side of the Ridge.

1

CONSTRUCTIVE / DIVERGENT BOUNDARY

MID-ATLANTIC RIDGE

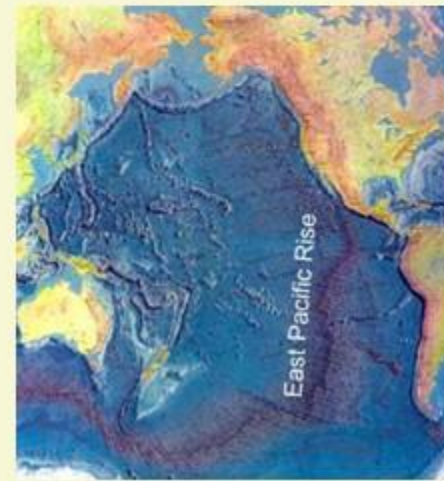
- 1** It runs down the Centre of the Atlantic Ocean.
- 2** It spreads apart at the rates of 2 to 5 cm per year.
- 3** The Ridge has a deep rift valley along its crest. The rift valley is 1 to 3 kms deep, about the depth and width of the Grand Canyon.



TWO OF THE MOST IMPORTANT MID-OCEAN RIDGES

EAST PACIFIC RISE

- 1** East Pacific Rise spreads fast at rates of 6 to 16 cm per year.
- 2** Due to the fast spreading rates there is no Rift Valley, just a smooth volcanic summit with a crack along the crest and this is much smaller than the Atlantic Rift Valley.





MID-ATLANTIC RIDGE

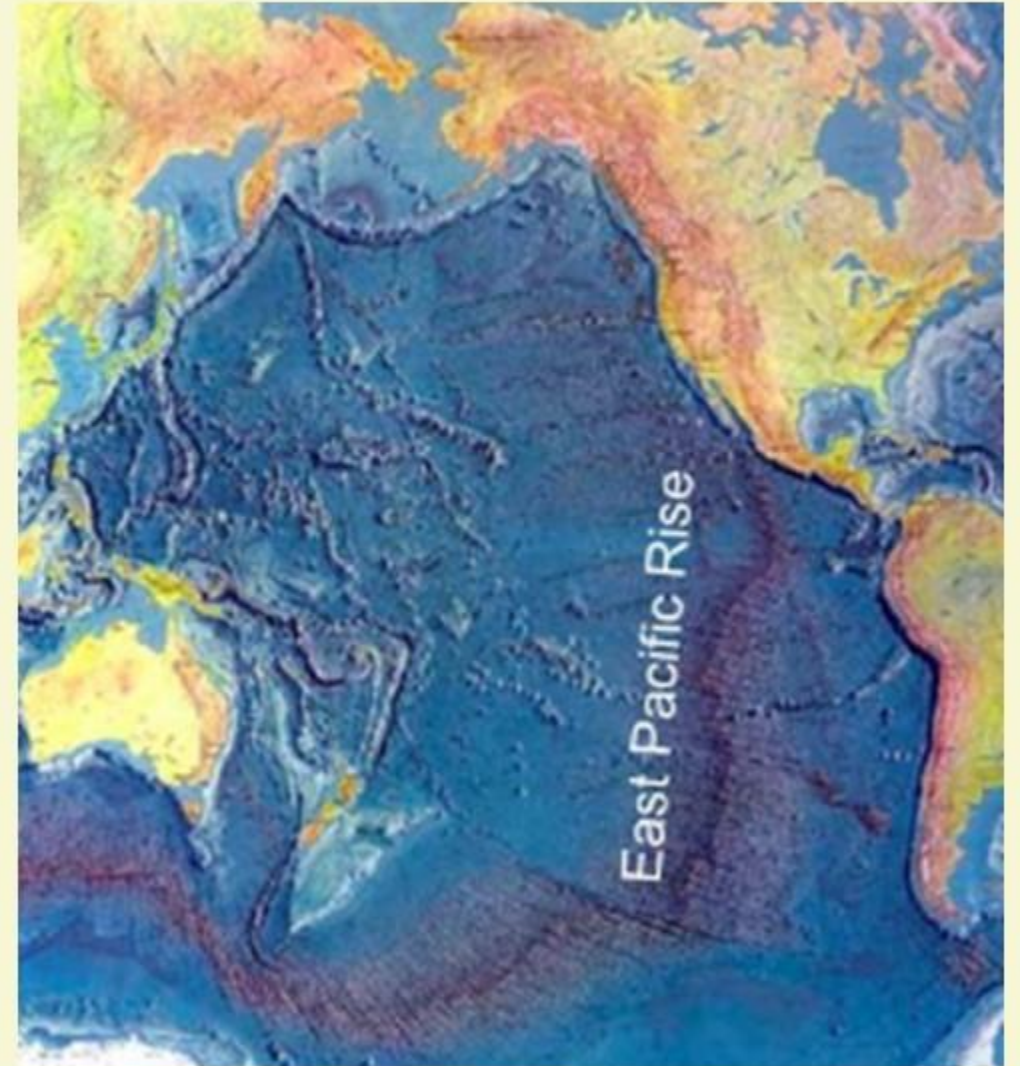
Jan Mayen (Norway)

Iceland

Azores (Portugal)

Ascension Island (UK)

Gough Island (UK)



1 CONSTRUCTIVE / DIVERGENT BOUNDARY

WHAT HAPPENS EXACTLY AT THE MID-OCEAN RIDGE?

1 Cracks allow the near freezing sea water to seep deep into the hot new crust.

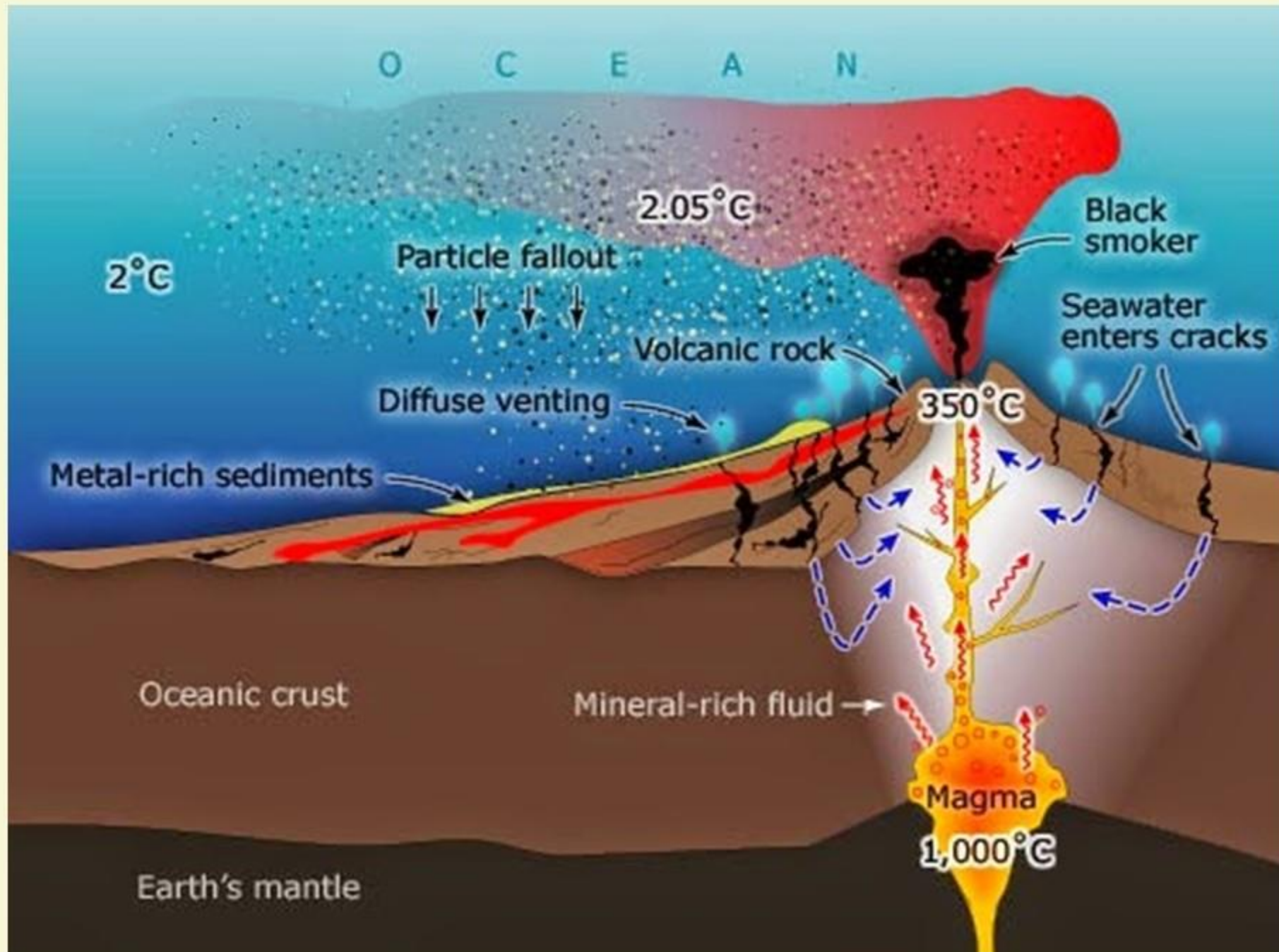
2 This water becomes superheated to temperatures greater than 400°C causing the water to become so buoyant and it shoots out of the seafloor at very high speeds. This looks like thick black smoke.

3 The hot hydrothermal water dissolves minerals out of the basalt crust and reacts with cold sea water, when coming back to the seafloor.

4 The black appearance is caused by precipitation of tiny mineral particles in the plume of hot water as it gushes out of the seafloor.

5 Mounds of minerals are deposited very rapidly around these "black smoker" hydrothermal vents.

6 The main minerals are zinc, copper and iron-sulphides, but these are not enough to make economic value because they are so deep.



HOW DOES IT BENEFIT THE DEEP-SEA ANIMALS?

- 1** The heat from the volcanoes and the hydrothermal vent waters provide the energy base for a very unusual community of deep sea animals.
- 2** These animals live completely independent of sunlight. They get their energy from the deep sea volcanoes.
- 3** Instead of relying on floral use of the sun's energy to make organic material through photosynthesis, the food chain relies on the microbes able to oxidize hydrogen sulphide and other compounds in hydrothermal vent waters.
- 4** They convert large amounts of carbon dioxide in sea water into organic material through a process called Chemosynthesis.
- 5** The Community consists of hundreds of species including clams, mussels, crabs, vent fish, octopus etc.



MID-ATLANTIC RIDGE

- 1** Mid-Atlantic Ridge is the largest geological feature on the planet.
- 2** It is mostly underwater mountain range in the Atlantic Ocean.
- 3** This runs from 87°N about 333 km south of the North Pole to sub-Antarctic Bourvet island at 54°S.
- 4** The Mid-Atlantic Ridge is about 3 km in height above the ocean floor and 1000 to 1500 km wide.
- 5** It has got several transform faults and an axial rift valley along its length.
- 6** The Ridge was discovered in 1950s.
- 7** It led to the theory of seafloor spreading and general acceptance of Wegener's theory of Continental Drift.
- 8** The Mid-Atlantic Ridge separates the North American Plate from the Eurasian Plate in the North Atlantic, and the South American Plate from the African Plate in the South Atlantic.
- 9** These plates are still moving apart, so the Atlantic is growing at the ridge, at a rate of about 2.5 cm per year in an east-west direction.



MID-ATLANTIC RIDGE

Jan Mayen (Norway)

Iceland

Azores (Portugal)

Ascension Island (UK)

Gough Island (UK)

HOW THE VOLCANIC ISLANDS ARE FORMED?

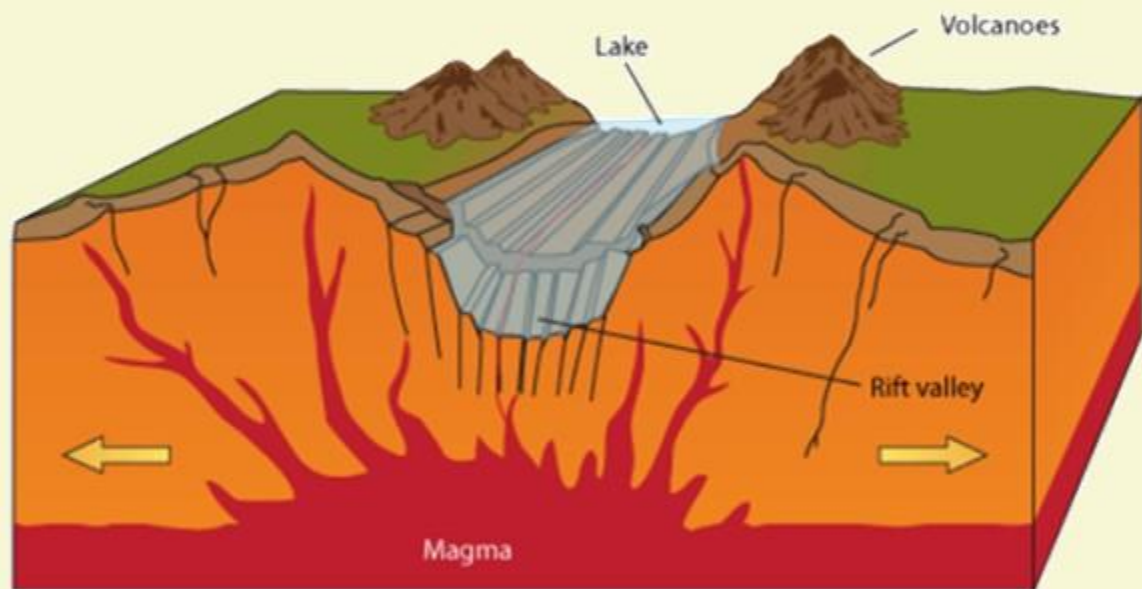


1 Most of the ridge system is under water but forms land as a set of volcanic islands of varying size that run the length of the Atlantic Ocean.

2 The Islands are ----

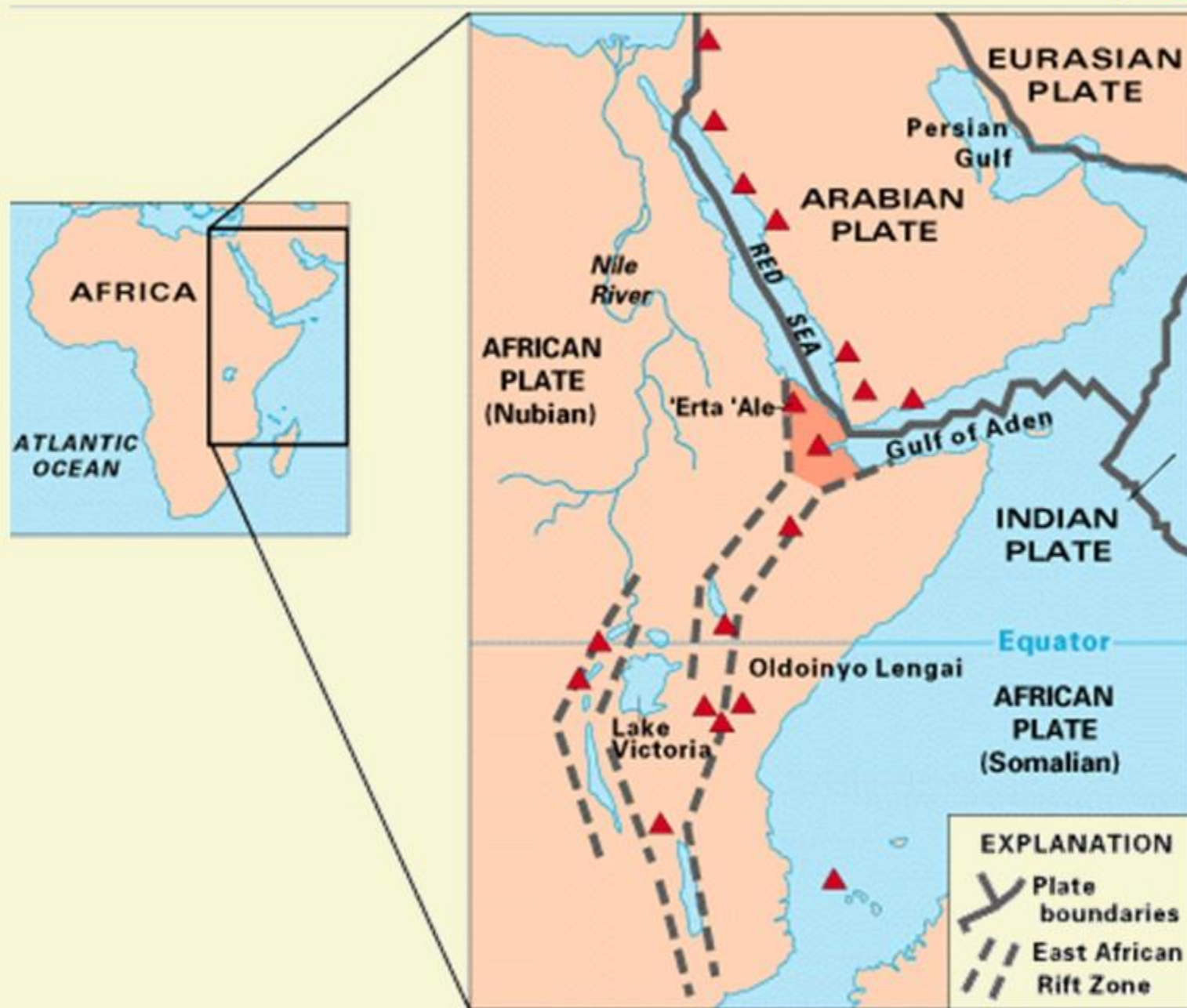
- Jan Mayen (Norway)
- Iceland
- Azores (Portugal)
- Ascension Island (UK)
- Gough Island (UK)

WHAT ABOUT THE CONTINENTAL RIFT VALLEYS?



- 1 These are the sites, where a continental landmass is ripping itself apart.
- 2 Africa will eventually split along the Great Rift Valley system.
- 3 Africa is now a single continent and it will emerge as two i.e., one on the African plate and the other on the smaller Somali plate.
- 4 The new Somali continent will be mostly oceanic, with the Horn of Africa and Madagascar its largest landmasses.

RIFT VALLEY AFRICA



WE EXPRESS OUR
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